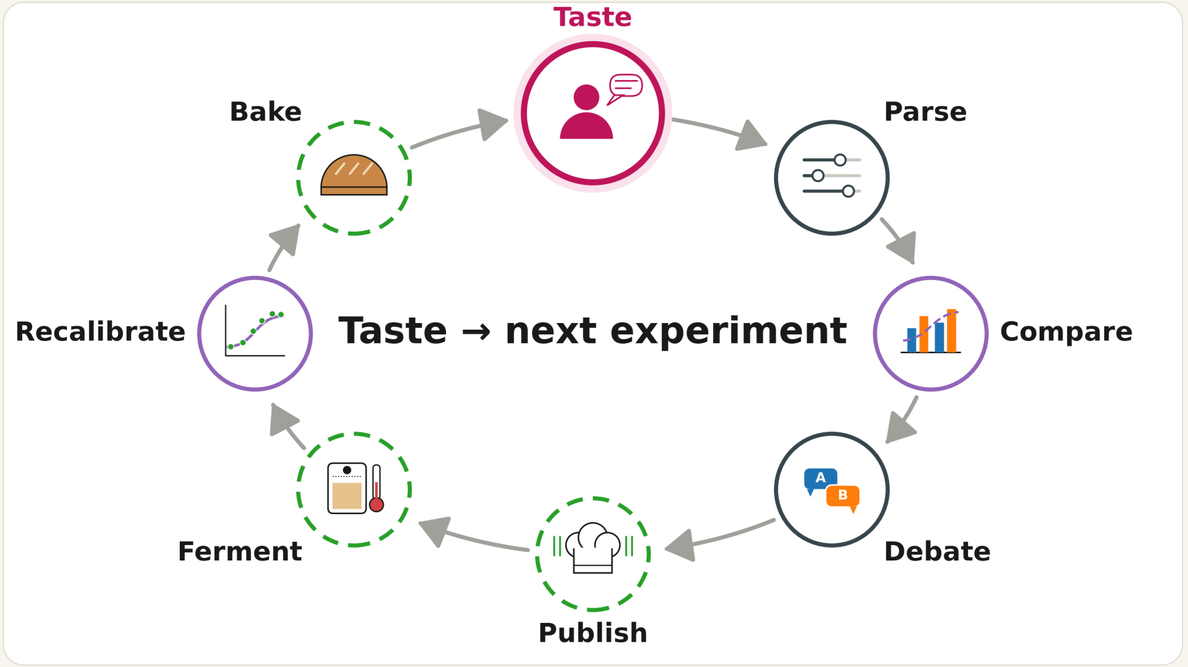
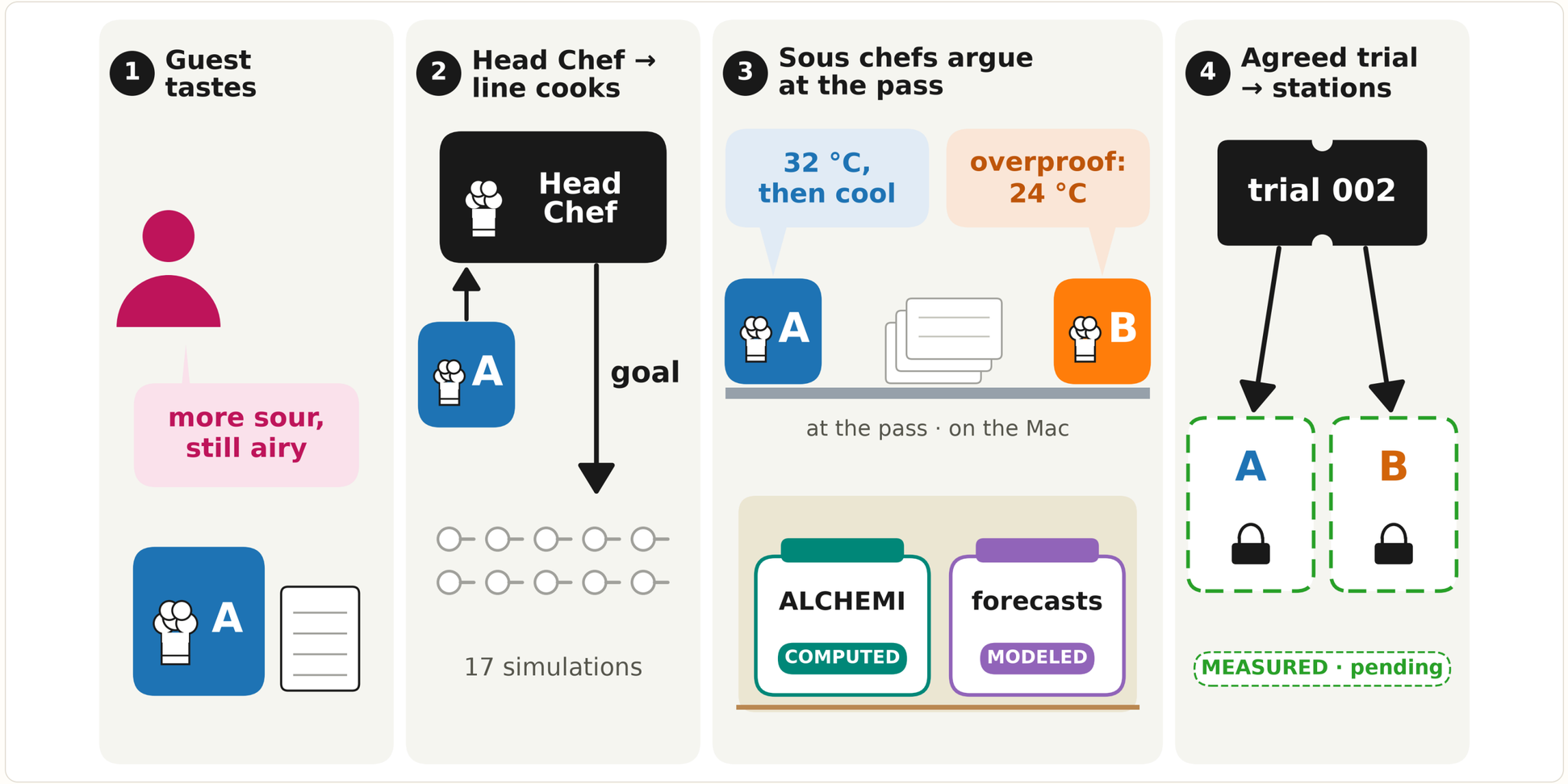
FERMENTAI · SCIENCECLAW 2026

Every kitchen can become a food-science lab

- **Problem:** flavour by guesswork
- **Vision:** every kitchen a lab
- **Why now:** cheap GPU chemistry + AI agents



A head chef, two sous chefs, two stations



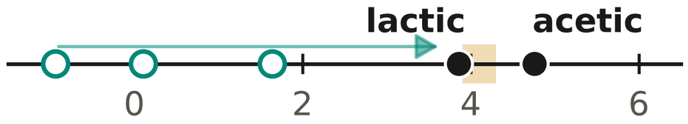
Two partners: NVIDIA chemistry, Formlabs fabrication

NVIDIA ALCHEMI

COMPUTED

which acid is stronger?

◄ stronger acid pKa weaker acid ►



GPU, more water → sourdough pH

1,066 GPU runs

Fermentation model

MODELED

recipe
schedule
starter
acid terms

ODE kinetics
yeast
bacteria
sugar
acids

pH
lactic/acetic
CO_2
rise
overproof



Formlabs PreForm API

COMPUTED

MODELED







2.4 h

239 mL

Clear V5

Working today, and what we need next

- **Computed:** GPU chemistry · print prep
- **Modeled:** forecasts · crumb · benchmark
- **Ask:** 3 pilot kitchens + GPU & printer credits

MODELED · in-silico

3.9 / 4 choices changed	**3.1 → 0** duplicate experiments
+0.68 nats more information	**same** final pick